

Gauss's Law

A powerful shortcut: relate the field on a closed surface to the charge inside it, and symmetric problems collapse to one line.

By **Pratik Bhanuse**, PhD Physics candidate, Western University · pratikbhanuse.github.io

Calculating fields by integrating Coulomb's law over a charge distribution (Chapter 13) works, but it is laborious. For distributions with high symmetry (spheres, long cylinders, flat sheets) there is a far more elegant route. **Gauss's law** relates the total electric “flow” through any closed surface to the charge enclosed. It is one of Maxwell's four equations, the foundations of all electromagnetism. Beyond calculating fields, it explains why charge on a conductor sits on its surface, why the field inside a metal box is zero, and why you are safe in a car during a lightning storm.

YOU WILL BE ABLE TO

- Calculate electric flux through flat and curved surfaces.
- State Gauss's law and explain its connection with Coulomb's law.
- Choose Gaussian surfaces to find fields for spherical, cylindrical and planar symmetry.
- Use Gauss's law for non-uniform charge distributions.
- Explain the properties of conductors in electrostatic equilibrium, shielding, and the field at a conductor's surface.

THINK FIRST

A sealed metal box is placed between the plates of a strong electric field. Inside the box is a sensitive electronic device. Will the device feel the external field?

No. Free electrons in the metal rearrange themselves within a tiny fraction of a second, until the field they produce exactly cancels the external field everywhere inside the metal *and* inside the empty cavity. This is a **Faraday cage**. MRI rooms are lined with copper for this reason, to keep outside radio signals out of the scanner. A car or aircraft body protects its occupants in a lightning strike in the same way, and coaxial cables shield the signals they carry.

§14.1

Electric flux

Picture the field as a flow. The **electric flux** through a surface measures how much field passes through it.

For a **flat surface** of area A in a uniform field, define the area vector $\vec{A}$: its magnitude is A and its direction is perpendicular to the surface. Then

$$\Phi_E = \vec{E} \cdot \vec{A} = EA \cos \theta,$$

where θ is the angle between $\vec{E}$ and the normal to the surface. The flux is greatest when the field passes straight through the surface ($\theta = 0$), and zero when the field skims along it ($\theta = 90^\circ$).

In general, split the surface into small patches $d\vec{A}$ and add up:

$$\Phi_E = \int \vec{E} \cdot d\vec{A} \quad (\text{SI unit: N m/C}).$$

For a **closed surface** we write $\oint$, and take $d\vec{A}$ to point **outward**. Flux leaving the surface is then positive, and flux entering is negative.

WORKED EXAMPLE 14.1**Flux through a cube and a tilted square**

(a) A cube of side a sits in a uniform field $\vec{E} = E\hat{i}$, with its faces perpendicular to the axes. Find the flux through each face and the total flux. (b) A square of side 10 cm lies in a uniform

field of 2.0×10^4 N/C, with its normal at 60° to the field. Find the flux through it.

Solution

(a) The face at the back ($-x$ side) has outward normal $-\hat{i}$, so its flux is $-Ea^2$. The front face has $+Ea^2$. The four side faces are parallel to $\vec{E}$ and have zero flux. The total is $\Phi = 0$: every field line that enters the cube also leaves it.

(b) $\Phi = EA \cos \theta = (2.0 \times 10^4)(0.010) \cos 60^\circ = 100 \text{ N m}^2/\text{C}$.

§14.2

Gauss's law

Take a point charge q at the centre of a sphere of radius r . The field is radial with magnitude kq/r^2 , so it is perpendicular to the sphere and has the same size everywhere on it:

$$\oint \vec{E} \cdot d\vec{A} = \frac{kq}{r^2} \cdot 4\pi r^2 = 4\pi kq = \frac{q}{\epsilon_0}.$$

The radius cancels, because the inverse-square fall in the field exactly matches the r^2 growth in area. Field lines neither start nor stop in empty space, so the same number of lines crosses *any* closed surface around the charge, whatever its shape. Lines from charges outside the surface enter and leave again, contributing zero net flux. Superposition then gives:

KEY IDEA

Gauss's law. The net electric flux through any closed surface equals the total charge enclosed divided by ϵ_0 :

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}.$$

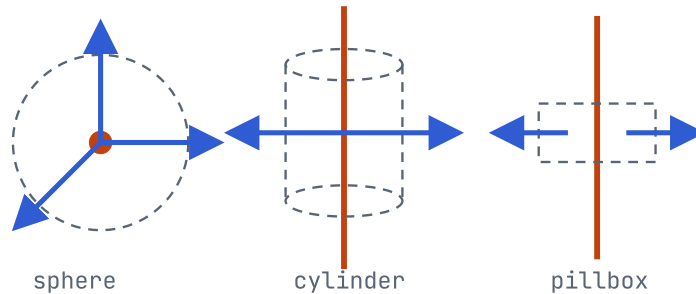
Charges outside the surface contribute nothing to the net flux, although they do affect the field at each point on it.

Gauss's law is always true. It is *useful* for finding $\vec{E}$ only when symmetry lets us pull E outside the integral. That needs a **Gaussian surface** on which E is either constant and perpendicular to the surface, or parallel to the surface so that it contributes no flux.

KEY IDEA

How to use Gauss's law.

1. Identify the symmetry: spherical, cylindrical or planar.
2. Choose a Gaussian surface that shares that symmetry and passes through the point where you want $\vec{E}$.
3. Evaluate $\oint \vec{E} \cdot d\vec{A}$, which becomes $E \times (\text{the relevant area})$.
4. Find Q_{enc} , integrating the charge density if necessary.
5. Solve for E .



Gaussian surfaces for the three symmetries: a sphere for a point or spherical charge, a coaxial cylinder for a line of charge, and a pillbox for a plane.

§14.3

Spherical symmetry

Thin spherical shell of radius R carrying charge Q . Use a concentric Gaussian sphere of radius r :

- Outside ($r > R$): $E \cdot 4\pi r^2 = Q/\epsilon_0$, so $E = kQ/r^2$. The shell acts like a point charge at its centre.
- Inside ($r < R$): $Q_{\text{enc}} = 0$, so $E = 0$.

These are the shell theorems of Chapter 8, now proved in one line each.

Uniformly charged solid sphere (charge Q , radius R , uniform density $\rho = Q/(\frac{4}{3}\pi R^3)$). Inside, the charge enclosed by a sphere of radius r is $Q(r/R)^3$:

$$E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0} \frac{r^3}{R^3} \Rightarrow E = \frac{kQ}{R^3} r \quad (r \leq R), \quad E = \frac{kQ}{r^2} \quad (r \geq R).$$

The field grows linearly inside, peaks at the surface, then falls as $1/r^2$. This is the same shape as the graph of $g(r)$ in Chapter 8.

WORKED EXAMPLE 14.2

A charged ball

An insulating sphere of radius 10 cm carries $5.0 \mu\text{C}$ spread uniformly through its volume. Find the field at 5.0 cm, 10 cm and 20 cm from its centre.

Solution

$$kQ = (8.99 \times 10^9)(5.0 \times 10^{-6}) = 4.50 \times 10^4 \text{ N m}^2/\text{C}.$$

- At 5.0 cm (inside): $E = \frac{kQr}{R^3} = \frac{4.50 \times 10^4(0.050)}{(0.10)^3} = 2.2 \times 10^6 \text{ N/C}.$
- At 10 cm (surface): $E = \frac{kQ}{R^2} = 4.5 \times 10^6 \text{ N/C}.$
- At 20 cm (outside): $E = \frac{4.50 \times 10^4}{0.040} = 1.1 \times 10^6 \text{ N/C}.$

The surface field is above the breakdown field of dry air (about $3 \times 10^6 \text{ N/C}$), so in practice the ball would discharge into the surrounding air through sparks.

§14.4

Cylindrical symmetry

Infinite line of charge with linear density λ . Use a Gaussian cylinder of radius r and length ℓ , coaxial with the line. The field is radial, so the flat end caps carry no flux. Over the curved side, E is constant and perpendicular:

$$E(2\pi r\ell) = \frac{\lambda\ell}{\epsilon_0} \Rightarrow E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2k\lambda}{r}.$$

Compare this one-line result with the integration in Problem P13.13. The same result holds outside any long, uniformly charged cylinder. Inside a cylindrical *shell*, the field is zero.

IN PRACTICE

Coaxial cable. A coaxial cable (for TV, internet or medical monitoring leads) has a central wire carrying charge $+\lambda$ per unit length, surrounded by a conducting shield carrying $-\lambda$. Gauss's law with a cylinder **outside** the shield encloses zero net charge, so the external field is **zero**: the signal's field is confined between the conductors and cannot interfere with nearby equipment. Run in reverse, the same argument shows that outside fields cannot get in. Between the conductors, $E = 2k\lambda/r$ is strongest at the surface of the inner wire. That sets the cable's voltage rating, since the insulation breaks down if the field there is too large (Problem P14.17).

§14.5

Planar symmetry

Infinite sheet with surface charge density σ . Use a pillbox (a short cylinder) that straddles the sheet, with end caps of area A . By symmetry, the field is perpendicular to the sheet and points away from it on both sides. Both end caps carry flux and the curved side carries none:

$$2EA = \frac{\sigma A}{\epsilon_0} \Rightarrow E = \frac{\sigma}{2\epsilon_0}.$$

The field is uniform: it does not depend on the distance from the sheet.

Two parallel sheets carrying $+\sigma$ and $-\sigma$: this is the parallel-plate capacitor of Chapter 16. Superpose the two sheet fields. Between the sheets they point the same way and add. Outside, they point opposite ways and cancel:

$$E_{\text{between}} = \frac{\sigma}{\epsilon_0}, \quad E_{\text{outside}} = 0.$$

WORKED EXAMPLE 14.3

Parallel plates

Two large parallel plates carry surface charge densities of $+3.0 \mu\text{C}/\text{m}^2$ and $-3.0 \mu\text{C}/\text{m}^2$. Find the field between them and outside them. What would the fields be if both plates carried $+3.0 \mu\text{C}/\text{m}^2$?

Solution

Opposite charges:

$$E_{\text{between}} = \frac{\sigma}{\epsilon_0} = \frac{3.0 \times 10^{-6}}{8.854 \times 10^{-12}} = 3.4 \times 10^5 \text{ N/C},$$

pointing from the positive plate to the negative one. Outside, $E = 0$.

Both positive: now the fields cancel *between* the plates, so $E = 0$ there. Outside they add, giving $3.4 \times 10^5 \text{ N/C}$ pointing away from the plates on both sides.

§14.6

Non-uniform charge distributions

When the charge density varies with position, integrate it to find Q_{enc} . The symmetry argument still works, provided the density depends only on r (or on the distance from the axis, or from the plane).

WORKED EXAMPLE 14.4

A sphere whose density grows outward

A sphere of radius R has charge density $\rho(r) = \rho_0 r/R$. Find the field inside it, and its total charge.

Solution

Enclosed charge. Add up thin shells of radius r' and volume $4\pi r'^2 dr'$:

$$Q_{\text{enc}}(r) = \int_0^r \rho_0 \frac{r'}{R} 4\pi r'^2 dr' = \frac{4\pi\rho_0}{R} \cdot \frac{r^4}{4} = \frac{\pi\rho_0 r^4}{R}.$$

Gauss's law:

$$E \cdot 4\pi r^2 = \frac{\pi\rho_0 r^4}{\epsilon_0 R} \Rightarrow E = \frac{\rho_0 r^2}{4\epsilon_0 R} \quad (r \leq R).$$

Total charge: $Q = \pi\rho_0 R^3$. Outside, $E = kQ/r^2$.

Check. At $r = R$, the inside formula gives $\rho_0 R/4\epsilon_0$, and $kQ/R^2 = \pi\rho_0 R/(4\pi\epsilon_0) = \rho_0 R/4\epsilon_0$. ✓ The field is continuous at the surface.

§14.7

Conductors in electrostatic equilibrium

A conductor contains charges that are free to move. When no current flows (electrostatic equilibrium), the following must be true.

KEY IDEA

Properties of a conductor in electrostatic equilibrium.

1. $\vec{E} = 0$ everywhere inside the material. If it were not, free charges would move, and the conductor would not be in equilibrium.
2. Any net charge resides on the surface. A Gaussian surface drawn just inside the material has $E = 0$ on it, so it encloses no charge.
3. Just outside, the field is perpendicular to the surface, with magnitude $E = \sigma/\epsilon_0$. A component along the surface would push charges around. A pillbox with only its outer face in the field gives $EA = \sigma A/\epsilon_0$.
4. A cavity with no charge inside it has $E = 0$ whatever the fields outside. This is shielding. If a charge q sits inside the cavity, a charge $-q$ gathers on the cavity wall, and the outer surface carries the conductor's net charge plus q .
5. Charge concentrates where the surface curves most sharply, so the field is strongest at points and edges.

WORKED EXAMPLE 14.5

A charge inside a conducting shell

A thick conducting spherical shell, with inner radius 5.0 cm and outer radius 8.0 cm, carries a net charge of $+4.0$ nC. A point charge of -2.0 nC sits at its centre. Find the charge on each surface of the shell, and the field at 3.0 cm, 6.0 cm and 10 cm from the centre.

Solution

Surface charges.

- A Gaussian sphere inside the metal ($5 < r < 8$ cm) has $E = 0$ on it, so it must enclose zero net charge. The inner surface therefore carries $+2.0$ nC, cancelling the -2.0 nC at the centre.
- The shell's total charge is $+4.0$ nC, so the outer surface carries $4.0 - 2.0 = +2.0$ nC.

Fields:

- At 3.0 cm: $Q_{\text{enc}} = -2.0 \text{ nC}$, so $E = \frac{(8.99 \times 10^9)(2.0 \times 10^{-9})}{0.030^2} = 2.0 \times 10^4 \text{ N/C}$, pointing inward.
- At 6.0 cm, inside the metal: $E = 0$.
- At 10 cm: $Q_{\text{enc}} = -2.0 + 4.0 = +2.0 \text{ nC}$, so $E = \frac{(8.99 \times 10^9)(2.0 \times 10^{-9})}{0.10^2} = 1.8 \times 10^3 \text{ N/C}$, pointing outward.

WORKED EXAMPLE 14.6**How much charge can a Van de Graaff hold?**

The spherical dome of a Van de Graaff generator has a radius of 15 cm. Air breaks down (sparks) when the field exceeds $3.0 \times 10^6 \text{ N/C}$. What is the maximum charge the dome can hold?

Solution

The field at the surface of the sphere is $E = kQ/R^2$, so

$$Q_{\text{max}} = \frac{E_{\text{max}} R^2}{k} = \frac{(3.0 \times 10^6)(0.15)^2}{8.99 \times 10^9} = 7.5 \times 10^{-6} \text{ C}.$$

The corresponding surface charge density is $\sigma = \epsilon_0 E = 2.7 \times 10^{-5} \text{ C/m}^2$.

Evaluate. A larger dome holds more charge, and reaches a higher voltage (Chapter 15), before it sparks. That is why research Van de Graaff accelerators have domes several metres across, often enclosed in pressurised insulating gas.

IN PRACTICE

Shielding and sharp points in practice.

- **Faraday cages.** MRI suites are lined with copper sheet or mesh to keep external radio-frequency noise out of the scanner's extremely sensitive receivers. Microwave ovens use a metal mesh in the door to keep their 12 cm waves inside, since the holes are far smaller than the wavelength. Aircraft and cars protect their occupants when struck by lightning, because the charge flows over the outer metal skin.
- **Sharp points.** The large fields at sharp points ionise the air, producing corona discharge. Lightning rods exploit this, and so do electrostatic precipitators and photocopier corona wires. The tips of high-voltage equipment are rounded to *prevent* corona, while static-discharge brushes on aircraft wings bleed off charge through deliberately sharp points.

§14.8

Gauss's law for gravity

Newton's gravity is also an inverse-square law, so it obeys a Gauss's law too. For the gravitational field $\vec{g}$:

$$\oint \vec{g} \cdot d\vec{A} = -4\pi G M_{\text{enc}}.$$

The minus sign appears because gravity is attractive. This gives the shell theorems and the linear rise of g inside a uniform Earth (Chapter 8) without any integration (Problem P14.16).

CHAPTER SUMMARY

- Flux: $\Phi_E = \int \vec{E} \cdot d\vec{A}$. For a flat surface in a uniform field, $EA \cos \theta$. For a closed surface, the outward normal is positive.
- Gauss's law: $\oint \vec{E} \cdot d\vec{A} = Q_{\text{enc}}/\epsilon_0$. It is always true, and useful when symmetry makes E constant over the surface.
- Sphere (shell or solid, outside): kQ/r^2 . Inside a shell: 0. Inside a uniform solid sphere: kQr/R^3 .
- Line or cylinder (outside): $\lambda/(2\pi\epsilon_0 r)$. Sheet: $\sigma/2\epsilon_0$. Two opposite sheets: σ/ϵ_0 between them and 0 outside.
- Conductors: $E = 0$ inside, net charge on the surface, $E = \sigma/\epsilon_0$ perpendicular to the surface just outside, shielded cavities, strongest fields at sharp points.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P14.1

The net electric flux through a closed surface is zero. Must the electric field be zero everywhere on the surface? Must the surface enclose no charge at all? Explain.

P14.2

Gauss's law is true for any closed surface. Why, then, can we use it to calculate the field of a uniformly charged sphere, but not that of a uniformly charged cube?

P14.3

Explain why the occupants of a car are protected if lightning strikes it. Would a convertible with a fabric roof offer the same protection?

P14.4

Excess charge is placed on a solid metal object. Where does it end up, and why? How is it distributed if the object is egg-shaped?

LEVEL B – STANDARD PROBLEMS**P14.5**

A hemisphere of radius 10 cm sits in a uniform field of 2.0×10^4 N/C that points along its axis of symmetry. Find the flux through the curved surface. (Hint: consider the closed surface formed by the hemisphere and its flat base.)

P14.6

A charge of 3.0 nC sits at the centre of a cube. Find the flux through each face. How would the answer change if the charge sat at one corner of the cube instead?

P14.7

A thin spherical shell of radius 0.30 m carries 6.0 nC. Find the field at 0.20 m and at 0.50 m from its centre.

P14.8

A long, solid insulating cylinder of radius 2.0 cm has a uniform charge density of $5.0 \mu\text{C}/\text{m}^3$. Find the field at 1.0 cm and at 4.0 cm from its axis. Comment on your answers.

P14.9

Two large parallel sheets carry $+3.0 \mu\text{C}/\text{m}^2$ and $-1.0 \mu\text{C}/\text{m}^2$. Find the field (magnitude and direction) in the three regions: to the left, between, and to the right of the sheets.

P14.10

A thick conducting spherical shell, with inner radius 4.0 cm and outer radius 6.0 cm, carries no net charge. A point charge of $+3.0$ nC sits at its centre. Find the charges on the inner and outer surfaces, and sketch $E(r)$ from $r = 0$ to $r = 10$ cm.

P14.11

In fair weather, there is a downward electric field of about 100 N/C near the Earth's surface. Treating the Earth as a conductor, find its surface charge density and its total charge.

P14.12

A Van de Graaff dome of radius 25 cm is to be operated at a surface field of 2.0×10^6 N/C, safely below breakdown. Find the charge on the dome and its surface charge density.

LEVEL C – CHALLENGE PROBLEMS**P14.13**

A sphere of radius R has charge density $\rho(r) = \rho_0(1 - r/R)$. (a) Find its total charge. (b) Find $E(r)$ inside and outside. (c) At what radius is the field strongest, and what is its maximum value?

P14.14

Cavity in a charged sphere. A uniformly charged insulating sphere (density ρ) contains a spherical cavity whose centre is displaced from the sphere's centre by a vector $\vec{a}$. Using superposition (sphere plus a “negative” sphere filling the cavity), show that the field inside the cavity is **uniform** and equal to $\rho\vec{a}/3\epsilon_0$.

P14.15

An infinite slab of thickness $2d$, between $x = -d$ and $x = +d$, has a uniform charge density ρ . Find the field inside and outside the slab, and sketch $E(x)$.

P14.16

Gravity also obeys a Gauss's law: $\oint \vec{g} \cdot d\vec{A} = -4\pi GM_{\text{enc}}$. Use it to show that inside a uniform planet of radius R and mass M , $g(r) = GMr/R^3$. Compare this with Chapter 8.

P14.17

Cable design. A coaxial cable has an inner conductor of radius 0.50 mm and an outer conductor of inner radius 3.0 mm. It carries ± 10 nC per metre. (a) Find the field at 1.0 mm from the axis, and the largest field anywhere in the cable. (b) The insulation between the conductors breaks down at 2.0×10^7 N/C. What is the maximum charge per metre the cable can carry? (c) Explain why the field outside the cable is zero, and why that matters for medical monitoring equipment.

P14.18

The hydrogen atom. In the ground state of hydrogen, the electron is described by a spherical “charge cloud” of density $\rho(r) = -\frac{e}{\pi a_0^3} e^{-2r/a_0}$, where $a_0 = 5.29 \times 10^{-11}$ m is the Bohr radius.

A proton sits at the centre. The charge of the cloud inside radius r is

$$-e \left[1 - e^{-2r/a_0} \left(1 + \frac{2r}{a_0} + \frac{2r^2}{a_0^2} \right) \right].$$

(a) Verify that the total charge of the cloud is $-e$. (b) Find the net charge inside $r = a_0$ and the electric field there. (c) Explain why a neutral hydrogen atom produces almost no field far away, even though the field inside it is enormous.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

© 2026 Pratik Bhanuse. All rights reserved. · pratikbhanuse.github.io